Pandemic Z: Optimizing Group Size for Survival in a Resource-Limited Zombie Outbreak

Leanne Gabrielle Igot
University of the Philippines Cebu
Cebu, Philippines
ldigot@up.edu.ph

Daenielle Rai Peladas
University of the Philippines Cebu
Cebu, Philippines
dapeladas@up.edu.ph

Jemimah Kaye Ronolo
University of the Philippines Cebu
Cebu, Philippines
jaronolo@up.edu.ph

Abstract

Traditional epidemiological models of zombie outbreaks often overlook the complexities of human social behavior, particularly how group dynamics influence survival outcomes. This study presents an agent-based simulation, Pandemic Z, which explores how varying group sizes affect survival in a resource-limited zombie apocalypse. Drawing from Realistic Conflict Theory, Social Identity Theory, and principles of Self-Organization, the model incorporates stress, trust, resource competition, and emergent cooperation among agents. Results suggest that group size significantly affects the outcomes of survival in a zombie-infested environment. Future research may delve deeper into enhancing realism where they can add more specialized human roles, environmental factors, and a deeper psychological system.

Keywords: *Agent-based Model, Zombie, Human, Agents, Theory, Zombie Apocalypse, Social behavior, Conflict*

1 Introduction

People often wonder how they would survive in extreme situations like natural disasters, pandemics, or even fictional scenarios such as a zombie outbreak. Zombie Apocalypse has become a popular metaphor for modeling crisis situations involving high uncertainty, resource scarcity, and social collapse. With the use of fictional crises, it allows researchers to explore how individuals and groups might behave under pressure. In any survival scenario, the ability to work effectively in a group can significantly influence outcomes. Group cooperation is a fundamental aspect of group dynamics, particularly relevant in situations that can be framed as social dilemmas, where individuals face a trade-off between acting selfishly for personal gain and cooperating for a shared group benefit [1].

While traditional epidemiological models like SIR have been adapted to simulate zombie outbreaks, they often rely on overly simplistic assumptions. These models typically treat populations as uniform and lack the nuance to capture dynamic human behavior, resource constraints, or decision-making under pressure, especially in a group setting [2]. As such, they fall short in realistically portraying survival scenarios where individual actions and group dynamics are critical.

In a post-apocalyptic, zombie-infested world where resources are scarce, human survival is critically dependent on group dynamics. While larger groups offer better defense and resource-gathering potential, they also increase competition for limited supplies. On the other hand, smaller groups may be more agile and less prone to resource shortages but might be more vulnerable to threats. Therefore, determining the optimal group size, the number of survivors that maximizes long-term survival chances, becomes a key problem.

This study aims to simulate a zombie outbreak scenario using an agent-based model in NetLogo to determine the optimal group size that will lead to the best survival outcome in a resource-limited environment. Specifically, this paper aims to: (1) Develop a NetLogo simulation that incorporates human, zombie, and shelter agents with survival-related variables (resources) and group behavior mechanisms; (2) Measure how group size and dynamics affect human survival, resource distribution, and interpersonal conflict; and (3) Identify optimal group size that lead to higher survival rates under varying environmental and resource constraints.

To ground the model in established theory, the simulation incorporates three key frameworks for human behaviors: **Realistic Conflict Theory**, which explains how competition for limited resources leads to group conflict; **Social Identity Theory**, which accounts for how group loyalty and trust influence cooperation; and **Self-Organization Principles**, which describe how roles and group coordination emerge naturally in decentralized systems. These theories guide the behavior of agents and shape how group dynamics are represented in the simulation.

2 Review of Related Literature

This study is grounded in three key theories—Realistic Conflict Theory (RCT), Social Identity Theory (SIT), and Self-Organization Principles—which help explain how individuals behave in groups under pressure, especially in scenarios involving scarce resources and external threats.

Realistic Conflict Theory (RCT), developed by Muzafer Sherif, explains that conflict arises when groups compete over limited resources. Sherif's Robbers Cave Experiment demonstrated that scarcity can lead to hostility and prejudice between groups. In this simulation, agents reflect this by competing for food and shelter. Larger groups experience more internal competition, while smaller ones often lack the manpower to survive, highlighting the trade-offs in group size [3, 4, 5].

Social Identity Theory (SIT), proposed by Henri Tajfel in the 1970s, focuses on how individuals derive a sense of self and esteem from group membership. People naturally categorize others into ingroups (trusted allies) and outgroups (potential threats). Repeated positive interactions, such as resource sharing or defending allies, foster trust and group cohesion [5]. This theory is reflected in the simulation through mechanisms that allow agents to form temporary alliances during zombie threats. Trust levels increase through cooperation.

Self-Organization Principles, rooted in systems theory and studies of emergent behavior, describe how complex group coordination can arise without any centralized control [6]. In both natural and social systems, local interactions among individuals often lead

to the spontaneous emergence of organized behavior. Agents adapt their behaviors in real time based on internal stress and surrounding threats. Some agents specialize in defense after repeated zombie encounters, while others focus on gathering resources.

Agent-based modeling is a methodology used for simulating systems composed of autonomous, interacting agents [7]. This approach has grown popular with the advancement of technology. Unlike traditional epidemiological models that treat populations as homogeneous mass, ABM can uniquely consider the spatial positioning of individual agents and zombies, as well as individual characteristics of humans to defend themselves and the abilities of individual zombies. Zombie outbreaks are often used as engaging examples to introduce ABM concepts.

In zombie-human agent-based models (ABMs), the primary agents are humans and zombies operating within a spatial environment, such as a grid or building. These simulations define specific behaviors for each agent, humans typically try to avoid zombies while zombies pursue humans. When agents are near each other or share the same space, interactions occur, such as zombies biting humans (possibly infecting them) or humans attempting to defend themselves [8]. These models seek to offer a more realistic representation of survival situations by accounting for individual behaviors and group interactions, going beyond the simplified assumptions of traditional models that treat populations as uniformly mixed [2].

While several studies explored zombie apocalypse using agent-based models, only a few have dug deeper into the complexities of group dynamics grounded in established social and behavioral theories, such as the work of Riebling and Schmitz (2016) who applied Norbert Elias' figurational sociology to model social dynamics of infection during a zombie outbreak [9]. The model explores three human strategies: (1) Escape, where humans cut ties with the zombies, (2) Sentimentality, where emotional bonds make cutting ties harder, and (3) Stigmatization, where humans sever connections with other humans with zombie contacts. This research explores how group strategies affect human survival and how social connections change during a zombie outbreak that shows social behaviour as a big role in spreading infection. It also highlights how using sociological ideas in simulations can help us understand real-life epidemics better.

Our study aims to integrate the established theories into an agent-based model which can help us understand how interactions between humans can be impactful in a challenging environment.

3 Methodology

This study used Agent-Based Modeling (ABM) approach using NetLogo to simulate human survival dynamics during a zombie apocalypse scenario, with the objective of investigating how dynamic role adoption affects survival rates, resource efficiency and social cooperation (measure in conflict frequency) over a simulated time period. The model focuses on emergent behavioral patterns where human agents adaptively specialize into survival roles based on their performance in combat and resource gathering activities.

3.1 Agent Design

The model utilizes three types of agents: humans, zombies and resources.

3.1.1 Human Agents

The human agents represented survivors in the post-apocalyptic environment. They are represented with the "person" shape in NetLogo with colors that change based on their specialized roles (white for generalist, blue for fighters, pink for gatherers). They are placed randomly in shelter locations during setup.

Human agents possessed multiple interconnected properties organized into several categories. Each human began with a set of **physical properties:** an energy level, representing their survival capacity that decreased with actions and could be restored by consuming resources. Speed determined movement capability, strength affected combat effectiveness, and vision range influenced detection of threats and opportunities. If a human becomes infected by a zombie, their visual representation would change to **purple**, signaling their transformation and growing danger to others.

In terms of **resource management**, humans maintained a resource inventory, representing their ability to carry and store essential supplies. The psychological state of each agent was shaped by three interdependent variables: stress, trust, and cooperation levels. Stress increased with zombie proximity and decreased in shelters (resting), while trust and cooperation levels affected willingness to share resources and assist other survivors.

A central feature of the model was the **dynamic role system** which introduced adaptive specialization among agents. Initially, all agents started as generalists possessing balanced combat skills and gathering skills. Over time, agents could specialize into one of two roles based on performance: fighters tend to focus on combat and defense while gatherers concentrate on resource collection while avoiding direct combat.

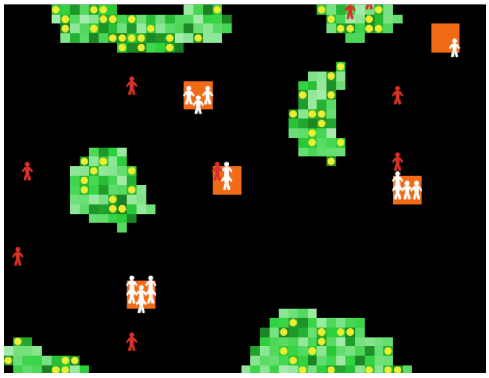

Figure 1: UI Interface of Project Z in Netlogo

Agent specialization was driven by **performance tracking,** where the model continually monitored the success-to-attempt ratios of combat actions, gathering task and defensive behaviors. Successful actions contributed to the accumulation of experience points, which in turn raised the agent's role confidence level on a scale from 0 to 100. Every 10 ticks, performance is evaluated (combat vs gatherer), if the difference between these two scores exceeded a period of 20 ticks was enforced between role transitions. This adaptive system ensures that agents developed roles suited to their demonstrated strengths, enabling more efficient survival strategies in a dynamic and hostile environment.

3.1.2 *Zombie Agent*

The zombie agents represented the primary threat in the simulation environment. They are represented with the "person" as well in NetLogo, colored red with variations indicating different states. They were placed randomly throughout the environment during setup.

Zombie agents [10] had basic survival properties including health, speed (0.2 slower than humans), strength for damage calculation, and sensory capabilities with vision range and hearing range for human detection. Zombies operated through a finite state machine with five distinct state, (a) Wandering, zombies walks around random movements searching for humans, (b) Chasing, direct pursuit of detected human targets, (c) Horde, joining other zombies in coordinated pursuits, (d) Feeding, consumes defeated humans to restore satiation, (e) Breaking-in, attempting to breach shelter defenses.

The simulation also featured several advanced zombie mechanics to enhance realism and challenge. Horde formation occurred when multiple zombies detected the same human, promoting collective pursuit behaviors. Shelter-breaking allowed zombies to erode barriers and compromise safe zones gradually. A key threat was the infection mechanic, where there was a 15% probability that a zombie attack would transmit the infection to a human, potentially turning them into an infected agent. Additionally, a decay system was implemented to simulate the deterioration of zombie agents over extended periods, reducing their effectiveness after simulated years of existence.

3.1.3 *Resource Agent*

Resource agents represented collectible supplies essential for human survival. They are represented with the "circle" shape in NetLogo, colored yellow. Resources spawned only in designated resource areas (green patches) based on environmental resource levels. Each resource agent contained an amount property (10-30 units) representing the quantity of supplies provided when collected by humans. Resources respawned in depleted areas based on environmental resource scarcity levels and patch-specific resource generation rates.

3.2 Environment Design

The model world represented a post-apocalyptic urban environment loaded from an external map file (map.txt) with dimensions corresponding to scaled urban areas. The environment was composed of three primary areas. Shelters, marked by orange patches, served as safe zones equipped with protective capabilities. Each shelter had an integrity value indicating its structural soundness, which could degrade over time due to repeated zombie assaults. Shelters not only provided physical protection but also reduced stress levels for any human agents inside, though they could eventually be breached through sustained attacks. Resource areas, shown as **green patches**, were designated locations for resource spawning and regeneration. Each of these areas had an associated resource level, which influenced both the probability of resource appearance and the rate at which they regenerated over time. These zones were critical for long-term survival, often drawing human agents into risky situations due to their exposure and vulnerability.

3.3 Experiment and Evaluation

The simulation experiment was designed to systematically investigate the relationship between group size and survival outcomes in a resource-constrained zombie apocalypse scenario. The experimental framework tested three distinct group size categories to capture the full spectrum of small group dynamics and their impact on survival effectiveness.

- **Small Groups (2-4 survivors):** These configurations tested minimal viable group sizes, examining whether reduced resource competition and simplified coordination could compensate for limited defensive capabilities and reduced task specialization.
- **Medium Groups (5-7 survivors):** This middle range explored the balance between having sufficient manpower for defense and resource gathering while maintaining manageable group coordination and resource distribution.
- **Large Groups (8-10 survivors):** These larger configurations investigated whether increased defensive strength and task specialization could overcome the challenges of heightened resource competition and complex group dynamics.

The experimental conditions were standardized across all group size variations to ensure valid comparisons. Each simulation maintained consistent environmental pressures with a fixed population of 8 zombies, creating a controlled threat level that remained constant regardless of human group size. A total of **50 simulation** runs *(150 per group size category)* were conducted across all group size configurations, providing sufficient statistical power to identify significant patterns while considering the stochastic nature of agent-based modeling. The runs were distributed proportionally across the group size ranges to ensure balanced representation of each category.

3.3.1 *Evaluation Metrics*

Data on survival rate, resource utilization efficiency, and conflict frequency were collected throughout the simulation to evaluate group performance under zombie threat. These metrics were tracked continuously to observe how different group dynamics and environmental conditions influenced human survival and behavior. Line plots and box plots were used to visualize trends across various group sizes and over time. To assess statistical significance, one-way ANOVA tests were conducted to determine whether differences in group structure, zombie density, or resource availability had meaningful effects on survival outcomes and social cohesion. This analytical approach helped validate the impact of key factors—such as zombie pressure and group adaptation—on the resilience and effectiveness of human agents in the hostile environment.

4 Results and Discussion

The statistical analysis revealed significant relationships between group size and all three primary outcome measures. Analysis of Variance (ANOVA) tests demonstrated that group size had a statistically significant effect on survival outcomes, resource management efficiency, and interpersonal conflict frequency, as summarized in **Table 1.**

Table 1: ANOVA Results for Group Size Effects on Key Metrics

Metric	F-Statistic	P-value
Survival Rate	16.1324	5.8483×10^{-21}
Resource Efficiency	22.4429	9.0165×10^{-29}
Conflict Frequency	6.6371	3.3143×10^{-8}

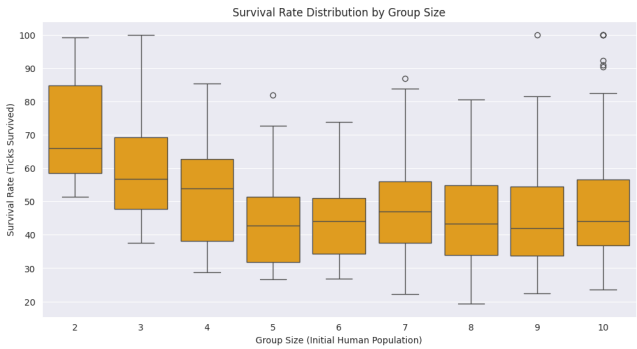

Figure 2: Box Plot of Survival Rates Across Group Sizes

All three metrics demonstrated highly significant relationships with group size ($p < 0.001$), providing strong statistical evidence that **group composition substantially influences survival dynamics** in resource-constrained environments.

4.1 Survival Rate Analysis

The analysis of survival rates across different group sizes revealed results that contradicted our initial hypothesis. Rather than medium-sized groups (5-7 members) demonstrating optimal survival rates, the data clearly indicated that smaller groups (2-4 members) achieved significantly higher survival outcomes.See Table 1.

Smaller groups performed better due to several key advantages. With fewer members, there was less competition for resources, giving individuals better access to supplies and reducing stress from scarcity. Their smaller size also allowed for easier movement within shelters and across the map, helping them reach resources quickly and avoid zombie threats. Less crowding meant a lower risk of infection, as members had more space and better shelter conditions. Overall, these factors created a less stressful environment, leading to better cooperation, smarter decisions, and more efficient use of resources.

4.2 Resource Utilization Efficiency

The analysis of resource utilization efficiency revealed a clear inverse relationship between group size and efficiency (Figure 2) as mentioned before. As group size increased, the efficiency of resource collection, distribution, and utilization decreased significantly.

Larger groups showed lower resource efficiency due to several factors. More members meant more foraging and longer exposure to threats, leading to higher energy use that reduced the net gain from collected resources. Increased competition raised stress levels, causing hoarding and poor sharing behavior. Coordination became

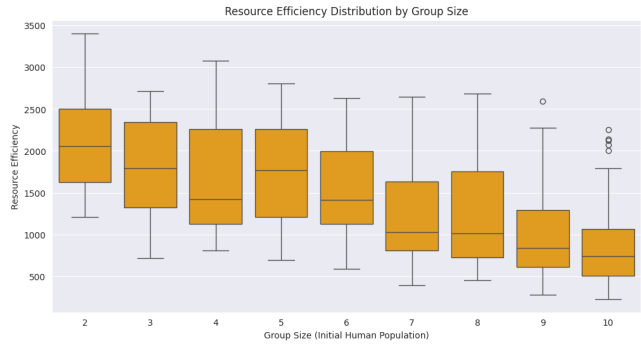

Figure 3: Resource Usage Efficiency Across Group Sizes

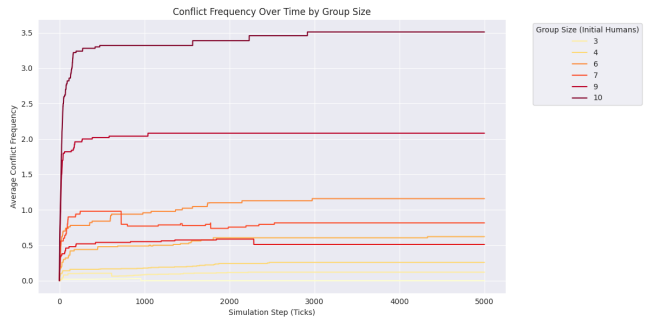

Figure 4: Timeline of Conflict Frequency by Group Size

harder, often resulting in overlapping efforts and missed resources. As group size grew, cooperation declined, and individuals became less willing to share—aligning with**Realistic Conflict Theory.** [3, 4].

4.3 Conflict Frequency Analysis

The analysis of interpersonal conflict revealed that smaller groups experienced significantly lower conflict rates compared to their larger counterparts (Figure 3). This finding supports the theoretical predictions derived from **Social Identity Theory and Realistic Conflict Theory** [3, 4, 6]

Smaller groups showed stronger social cohesion due to several key factors. With fewer interactions, there was less chance for conflict, making relationships easier to manage. More available resources reduced competition and tension, while better access to shelter lowered stress and improved mental well-being. These conditions helped members cooperate more and build stronger bonds. The close-knit setting also made it easier to develop trust, encouraging mutual support and teamwork.

4.4 Emergent Behavioral Patterns

The simulation demonstrated how human agents adapted their roles based on repeated exposure to specific environmental conditions. Those frequently engaging with resource zones improved their gathering abilities and gradually specialized as gatherers, while agents encountering frequent zombie threats enhanced their

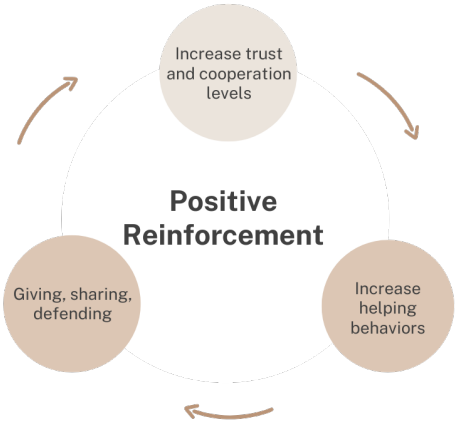

Figure 5: Positive Reinforcing Cycle

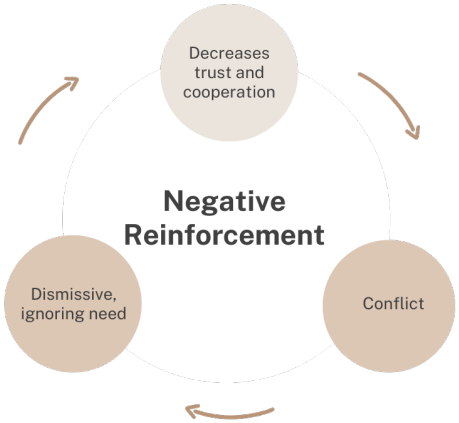

Figure 6: Negative Reinforcing Cycle

combat proficiency and transitioned into defenders. This adaptive behavior allowed agents to align their roles with the demands of their surroundings. The role specialization system was highly responsive to environmental pressures. Agents regularly assessed their success in combat and gathering, adjusting their focus as needed. Agents displayed patterns of social grouping based on shared traits and trust levels. Gatherers often **clustered in resource-rich zones**, suggesting an inherent tendency for collaboration among individuals with similar goals. This trait-based clustering supported efficient resource collection and minimized overlap in task execution. Additionally, trust and cooperation formed a reinforcing cycle—high-trust relationships encouraged cooperative actions like sharing and defense, while positive interactions further deepened mutual trust.

Agents displayed notable patterns of social grouping based on shared traits and trust levels. Gatherers often clustered in resource-rich zones, suggesting an inherent tendency for collaboration among individuals with similar goals. This trait-based clustering supported efficient resource collection and minimized overlap in task execution. Additionally, trust and cooperation formed a reinforcing cycle—high-trust relationships encouraged cooperative actions like sharing and defense, while positive interactions further deepened mutual trust. See Figure 5.

Access to shelters played a critical role in managing group stress and maintaining cohesion. Shelter spaces consistently lowered stress levels, reducing the likelihood of conflicts and encouraging more collaborative behaviors. Successful teamwork, such as assistance during attacks or resource shortages, further reinforced skill growth and trust, contributing to long-term group resilience. However, the absence of support or repeated dismissive behavior weakened social bonds. Agents who ignored the needs of others saw a decline in trust and cooperation from group members, increasing the risk of fragmentation (Figure 6). This highlighted the importance of mutual support and positive reinforcement in sustaining group unity and collective survival.

5 Limitations

The model did not incorporate family or kinship bonds, which in real-world scenarios often play a significant role in influencing cooperation, trust, and resource-sharing behaviors. Also the role system was limited to three categories—generalist, fighter, and gatherer. While useful for modeling adaptive specialization, this structure does not capture the full complexity of real-world survival roles, such as medics, engineers, or leaders, which could impact group performance and strategy. Lastly, the simulation lacked a detailed disease or medical system beyond the zombie infection mechanic. Realistic post-apocalyptic scenarios would likely involve illnesses, injuries, and medical care as critical survival factors. However, it's important to note that implementing such systems in **NetLogo was limited** by both its simplicity and the technical difficulty of layering additional mechanics without compromising performance.

6 Conclusion

This study successfully demonstrated that group size significantly affects survival outcomes in resource-constrained crisis scenarios, though not in the initially hypothesized manner. Through agent-based modeling incorporating established social theories (Realistic Conflict Theory, Social Identity Theory, and Self-Organization Principles), the research revealed that **smaller groups (2-4 members) consistently outperformed larger groups** across all measured outcomes. The key findings challenge strength in numbers during crisis situations. Smaller groups achieved higher survival rates ($F = 16.13$, $p < 0.001$), resource utilization efficiency ($F = 22.44$, $p < 0.001$), and significantly lower conflict frequencies ($F = 6.64$, $p < 0.001$). These advantages stemmed from reduced resource competition, enhanced mobility, lower stress levels, and stronger social cohesion. The simulation revealed sophisticated emergent behaviors, including **dynamic role specialization** where agents adapted to become either fighters or gatherers based on environmental exposure, and **trust-cooperation feedback** loops that strengthened group resilience. Crucially, the study demonstrated that access to communal

safe spaces and positive reinforcement mechanisms were essential for maintaining group stability. **Implications could extend beyond fictional scenarios** to real-world crisis management, suggesting that smaller, well-coordinated teams may be more resilient than larger groups in resource-scarce emergency situations. This insight could inform disaster preparedness strategies, emergency response protocols, and community resilience planning.

7 Recommendations

To enhance realism and depth, future versions of the simulation should include more human roles such as medics, scouts, or leaders to reflect real-world specialization. Environmental factors like rain, cold, fog, and time of day should be added to affect agent behavior, along with varied terrain features such as elevations, rivers, and forests to influence movement and strategy. Zombies can be made more adaptive by allowing them to use resources for survival, making them a longer-lasting threat. Introducing multiple human groups or factions could simulate alliances or conflicts under stress. Lastly, improving the psychological system to track individual relationships would allow trust and cooperation to develop based on personal interactions, adding depth to social dynamics.

References

[1] Jessica L Barker, Pat Barclay, and Hudson K Reeve. "Competition over personal resources favors contribution to shared resources in human groups". In: *PLOS ONE* 8.3 (2013), e58826. DOI: 10.1371/journal.pone.0058826.

[2] N. Smith. *What are your chances during a zombie apocalypse?* Accessed: 2025-06-02. Open Learning. n.d. URL: https://www.open.edu/openlearn/digital-computing/what-are-your-chances-during-zombie-apocalypse.

[3] Julie. *Realistic Conflict Theory: Psychology Definition, History & Examples.* Accessed: 2025-06-02. Dr. Philip G. Zimbardo. Dec. 2023. URL: https://www.zimbardo.com/realistic-conflict-theory-psychology-definition-history-examples/.

[4] *Social Psychology Course.* Accessed: 2025-06-02. Study.com. n.d. URL: https://study.com/academy/course/social-psychology.html.

[5] Administrator. *Realistic Group Conflict Theory in Social Psychology - iResearchNet.* Accessed: 2025-06-02. iResearchNet. Jan. 2016. URL: https://psychology.iresearchnet.com/social-psychology/social-psychology-theories/realistic-group-conflict-theory/.

[6] ScienceDirect. *Self-Organization.* 2025. URL: https://www.sciencedirect.com/topics/social-sciences/self-organization (visited on 06/02/2025).

[7] 4CDA. *An Introduction to Agent-Based Modelling.* 2025. URL: https://4cda.com/an-introduction-to-agent-based-modelling/ (visited on 06/02/2025).

[8] Charles M. Macal. "Tutorial on Agent-Based Modeling and Simulation: ABM Design for the Zombie Apocalypse". Presented at a simulation conference. 2018. URL: http://simulation.su/uploads/files/default/2018-macal-1.pdf (visited on 06/02/2025).

[9] Jan R. Riebling and Andreas Schmitz. "ZombieApocalypse: Modeling the Social Dynamics of Infection and Rejection". In: *Methodological Innovations* 9 (2016). DOI: 10.1177/2059799115622767.

[10] Max Brooks. *The Zombie Survival Guide: Complete Protection from the Living Dead.* New York: Three Rivers Press, 2003. ISBN: 9781400049622.

8 Author's Note

Leanne Gabrielle, Daenielle Rai, and Jemimah Kaye all contributed from initial brainstorming to model development and the writing of the paper. The development of the model was led by Daenielle Rai Peladas. The paper was written by Leanne Gabrielle and Jemimah Kaye, and was proofread by Daenielle Rai.